
BIOLOGY

9700/42

Paper 4 A Level Structured Questions

October/November 2017

MARK SCHEME

Maximum Mark: 100

Published

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Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

Question	Answer	Marks
1(a)(i)	<i>three from:</i> 1 the numbers of, prey / herbivores / sheep / goats, would increase ; 2 decrease in number of plants / <i>idea of</i> overgrazing ; 3 damage to habitat ; e.g. erosion 4 increased competition (for plants) ; 5 lack of food causes, herbivore numbers to decline / herbivores to migrate away ; 6 reduction in, biodiversity / species diversity / genetic diversity / genetic variation ; 7 AVP ; e.g. increase in other predators / disruption to food web	3
1(a)(ii)	<i>one from:</i> 1 snow leopards are, hard to see / camouflaged ; 2 vast habitat to cover (to locate snow leopards) / inaccessible locations / solitary ; 3 AVP ; e.g. difficulty getting permission from some governments to go into certain areas	1

Question	Answer	Marks
1(b)	<i>three from:</i> 1 ban trade (of, leopard / leopard products) ; 2 ban hunting ; 3 detection methods ; e.g. customs / trading officers / wildlife forensics / extra rangers 4 enforcement measures ; e.g. fines / prison / punishment 5 raise public awareness / education ; 6 <i>ref. to</i> protected areas / national parks ;	3

Question	Answer	Marks
1(c)	<i>two from:</i> 1 genetic diversity decreases / AW ; 2 increased chance of two harmful recessive alleles coming together / increase in inbreeding depression / decrease in hybrid vigour ; 3 (some) less likely to survive ; 4 (because) less likely to adapt to change in, environmental conditions / selection pressure ;	2
1(d)	<i>two from:</i> 1 appearance / morphology / visible features / looked similar to leopards ; 2 breeding (to produce fertile offspring) ; ora 3 <i>ref. to</i> behaviour ;	2
2(a)(i)	NADP ;	1
2(a)(ii)	<i>two from:</i> 1 photolysis of water / AW ; 2 <i>ref. to</i> photosystem II ; 3 using an enzyme ; 4 hydrogen ions and electrons combine to form hydrogen / AW ;	2

Question	Answer	Marks
2(b)(i)	<i>two from:</i> 1 to, reduce / stop, activity of enzymes ; 2 <i>ref. to</i> proteases / lipases ; 3 to prevent damage (to chloroplasts) ;	2
2(b)(ii)	1 to, control pH / keep pH constant ; 2 so enzyme works at optimum / to prevent denaturation of enzyme ;	2
2(b)(iii)	1 to avoid, osmosis / movement of water down a water potential gradient ; 2 to prevent damage (to chloroplasts) / AW ;	2
2(c)(i)	43 / 43.0 ;	1
2(c)(ii)	23.3 / 23.0 / 23 ; <i>accept ecf from 2(c)(i)</i>	1

Question	Answer	Marks
2(c)(iii)	<i>four from:</i> <i>relationship</i> 1 as light intensity increases time taken for decolourisation decreases or as light intensity increases rate of decolourisation increases or as light intensity increases decolourisation is faster ; <i>explanation (max 3)</i> 2 more, photons / <u>light energy</u>, absorbed ; 3 more / faster, photolysis / AW ; 4 more electrons, excited / released or faster rate of electron release ; 5 more protons released or faster rate of proton release ; 6 more / faster, <u>reduction</u> of DCPIP ;	4

Question	Answer	Marks
3(a)	<i>two from:</i> 1 killing by humans / eaten by humans / hunting / poaching ; 2 loss of, habitat / waste land ; 3 loss of, (plant) food / grazing ;	2
3(b)	<i>three from:</i> 1 DNA / nucleotide, sequencing ; 2 <i>ref. to</i> mitochondrial DNA ; 3 protein / amino acid, sequencing ; 4 <i>idea of</i> comparing sequences (from A and B) ; 5 AVP ; e.g. sequencing method such as Sanger / dideoxynucleotides / chain termination / automated / next generation or <i>ref. to</i> bioinformatics / genome analysis / microarray analysis	3

Question	Answer	Marks
3(c)	<i>three from:</i> 1 A (originally) geographically isolated / described, (from ancestors of, B / C / D) ; 2 for, 12 000 years / long period of time / many generations ; 3 different mutations (in Ireland and Great Britain) ; 4 different, selection pressures / environmental conditions, (in Ireland and Great Britain) ; 5 <i>ref. to</i> genetic drift ; 6 reduction in genetic diversity in A / population bottleneck ; 7 AVP ; e.g. no gene flow between populations	3

Question	Answer	Marks
4(a)	<i>four from:</i> <i>vasodilation</i> 1 arterioles / blood vessels, (in skin) widen ; R capillaries I arteries 2 more blood flows to capillaries / blood flows through capillaries more slowly ; 3 heat (energy) lost ; <i>sweating</i> 4 more sweat, produced / released ; 5 sweat / water, evaporates ; 6 using heat energy / <i>ref. to</i> latent heat of vaporisation ;	4
4(b)	<i>four from:</i> 1 high blood glucose concentration / hyperglycaemia ; 2 (causes) decrease in water potential (of blood) ; 3 detected by, osmoreceptors / hypothalamus ; 4 feelings of thirst ; 5 less / no, glucose converted to, fat / glycogen ; 6 glucose lost in urine / not all glucose reabsorbed / blood glucose concentration above the renal threshold ; 7 glucose not taken up by cells ; 8 (so) fats / proteins, are, respired / metabolised ;	4

Question	Answer	Marks
4(c)(i)	<i>three from:</i> 1 (pad contains) glucose oxidase ; 2 enzyme / glucose oxidase, reacts with glucose (in the blood) ; 3 oxygen detected ; 4 (electric) current generated ; 5 detected by electrode ; 6 gives numerical value (of blood glucose concentration) ;	3
4(c)(ii)	<i>one from:</i> 1 gives the, actual / accurate / exact, reading (of blood glucose concentration) ; 2 re-usable ; 3 more precise reading / quantitative ;	1
5(a)(i)	<i>two from:</i> 1 CAG / repeat, is a triplet or triplets of bases / codons, are added ; 2 (so) other triplets are unchanged ; 3 (so) no frameshift ;	2

Question	Answer	Marks
5(a)(ii)	<i>four from:</i> 1 allele is dominant ; 2 so will still be expressed even when, normal / recessive, allele is present ; ora 3 (gene therapy only) used to treat recessive (allele) disorders ; 4 cannot, remove dominant allele / replace an allele ; 5 dominant allele affects tissues in many parts of the body ;	4
5(b)(i)	50% / 1 in 2 / 0.5 / half ; A 100% if candidate implies parent is homozygous dominant	1
5(b)(ii)	<i>two from:</i> <i>advantage (max 1)</i> 1 can choose whether to have children ; 2 can prepare for the future ; 3 (if negative) removes anxiety / AW ; <i>not awarded with mp5</i> <i>disadvantage (max 1)</i> 4 (if positive) no treatment possible ; 5 (if positive) may lead to anxiety / AW ; <i>not awarded with mp3</i> 6 (if positive) social / financial, discrimination ; e.g. life insurance refusal 7 (if positive) may still not develop disease (<i>ref. to below 39 repeats</i>) ;	2
5(c)(i)	removing a cell from an embryo (for testing) ;	1
5(c)(ii)	to amplify the, DNA / gene, (from the embryo cell) ;	1

Question	Answer	Marks														
5(c)(iii)	<i>two from:</i> 1 embryos might be destroyed ; 2 wrong for parents to choose / designer embryos ; 3 contrary to, beliefs / values ; 4 less chance of Huntington allele being passed on / decrease in frequency of Huntington allele ; 5 <i>idea that</i> people with the faulty allele who otherwise would not have children can now do so ;	2														
6(a)(i)	1 as the eugenol concentration increases the percentage decrease in sodium ion movement increases ; 2 percentage at two concentrations of eugenol plus units ; <table><tr><th>conc (mmol dm⁻³)</th><th>%</th></tr><tr><td>0.0</td><td>0</td></tr><tr><td>0.2</td><td>30</td></tr><tr><td>0.4</td><td>50</td></tr><tr><td>0.6</td><td>65</td></tr><tr><td>0.8</td><td>72</td></tr><tr><td>1.0</td><td>80</td></tr></table>	conc (mmol dm ⁻³)	%	0.0	0	0.2	30	0.4	50	0.6	65	0.8	72	1.0	80	2
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6(a)(ii)	57–58 (%) ;	1														

Question	Answer	Marks
6(b)	<i>five from:</i> 1 reduced entry of Na^+ into sensory neurone ; 2 eugenol prevents opening of sodium ion channels / described ; 3 no / reduced, depolarisation of sensory neurone <u>membrane</u> ; 4 receptor potential / threshold potential, not reached ; 5 no / fewer, action potentials / impulses ; 6 may affect sodium-potassium pump ; 7 resting potential not restored ; 8 action potentials / impulses, do not reach brain ;	5
7(a)	1 inorganic phosphate added to ADP / $\text{ADP} + \text{P}_i$; 2 <i>ref. to</i> (named) phosphorylated compound ;	2
7(b)(i)	<u>ester</u> ;	1
7(b)(ii)	carrier of, 2C (unit) / acetyl group / acetate, to, the Krebs cycle / oxaloacetate ;	1
7(b)(iii)	<i>two from:</i> 1 more C–H (bonds) / more reduced / more hydrogen ; 2 produces more reduced NAD ; 3 more, aerobic respiration / oxidative phosphorylation / chemiosmosis ; 4 produces more ATP per, gram / unit mass ;	2

Question	Answer	Marks
7(c)(i)	17 ;	1
7(c)(ii)	0.71 ;; <i>one mark for $12 \div 17$</i> <i>allow ecf from (i) for one mark</i>	2
8(a)	<i>two from:</i> 1 (two or more) genes on same chromosome ; 2 not sex chromosome ; 3 will be inherited together ;	2
8(b)	<i>four from:</i> 1 most offspring are, purple long and red round / parental phenotypes ; 2 fewer offspring are, purple round and red long / non-parental phenotypes ; 3 recombinants / new combinations of alleles ; 4 (because of) crossing over ; 5 between the two gene loci ; 6 AVP ; e.g. COV is 12% / low COV / two genes not very far apart	4

Question	Answer	Marks
8(c)	<i>three from:</i> 1 change in, base (pair) / nucleotide, sequence in <u>DNA</u> ; 2 random / spontaneous ; <i>due to</i> 3 <i>ref. to</i> base, substitution / deletion / insertion ; 4 (during) DNA replication ; 5 failure of proof reading mechanism ; 6 <i>ref. to</i> mutagenic agents ; e.g. ionising radiation / UV radiation	3

Question	Answer	Marks
9(a)	<i>eight from:</i> 1 double membrane ; <i>inner membrane</i> 2 folded / cristae ; 3 increased / large, surface area ; 4 has, ATP synthase / stalked particles ; 5 has, carrier (proteins) / cytochromes ; 6 (site of) ETC / oxidative phosphorylation / chemiosmosis ; <i>intermembrane space</i> 7 has low pH / high concentration of protons ; 8 accepts protons from ETC / AW ; 9 proton gradient between intermembrane space and matrix or protons move from intermembrane space to matrix ; 10 <i>ref. to</i> ATP synthesis ; <i>matrix</i> 11 contains enzymes ; 12 site of, link reaction / the Krebs cycle ; <i>outer membrane</i> 13 presence of carriers for, pyruvate / reduced NAD ; 14 AVP ; e.g. ribosomes / DNA, plus function	8

Question	Answer	Marks
9(b)	<i>seven from:</i> <i>liver cells (max 6)</i> 1 only glycolysis occurs ; 2 pyruvate, cannot enter mitochondrion / remains in the cytoplasm ; 3 (pyruvate) becomes, hydrogen acceptor / reduced ; 4 by reduced NAD (from glycolysis) ; 5 lactate produced ; 6 lactate dehydrogenase ; 7 production of, 4 ATP / 2 ATP / small amount of ATP ; 8 allows glycolysis to continue ; <i>yeast cells</i> 9 decarboxylation / CO₂ removed ; 10 ethanal becomes, hydrogen acceptor / reduced ; 11 two steps (instead of one) ; 12 irreversible reaction (instead of reversible) ; 13 ethanol dehydrogenase ; 14 ethanol produced ;	7

Question	Answer	Marks
10(a)	<i>eight from:</i> 1 chiasmata formation ; 2 crossing over ; 3 exchange of genetic material ; 4 between non-sister chromatids of homologous chromosomes ; 5 at prophase 1 ; 6 new combination of alleles ; 7 linkage groups broken ; 8 independent / random, assortment, of homologous pairs / bivalents / AW ; 9 in metaphase 1 ; 10 2^n combinations / very large number of combinations ; 11 independent / random, assortment of, chromatids / chromosomes (correct context) ; 12 in metaphase 2 ; 13 <i>ref. to</i> mutation ;	8

Question	Answer	Marks
10(b)	<i>seven from:</i> 1 <i>ref. to TYR</i> gene ; 2 normal gene product is tyrosinase ; 3 tyrosine converted to, DOPA / dopaquinone ; ora 4 melanin / pigment, made ; ora 5 in melanocytes ; 6 mutant allele is recessive ; 7 tyrosinase, not produced / inactive ; 8 affects, hair / skin / irises ; 9 only in homozygous recessive people ;	7